Studying Ground-State Energy Estimation Through Quantum Sampling

Qiskit Fall Fest 2026

Team Mensia

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October 2026

Problem and Motivation

- Ground-state energy is an important quantity in quantum physics and chemistry.
- Exact diagonalization becomes expensive as the number of qubits increases.
- Sample-based quantum diagonalization (SQD) uses measured computational-basis states to construct a smaller subspace.
- The reduced Hamiltonian is then diagonalized using classical numerical methods.

Objective: Study how the number of measurement samples affects ground-state energy estimation.

Model and Qiskit Tools

We consider the transverse-field Ising chain:

$$H = -\sum_{i=0}^{N-2} Z_i Z_{i+1} - g \sum_{i=0}^{N-1} X_i.$$

Experimental parameters

- Number of qubits: $N = 8$
- Transverse-field strength: $g = 0.4$
- Open boundary conditions

Software tools

- `SparsePauliOp`: Hamiltonian representation in Qiskit.
- `StatevectorSampler`: generates computational-basis measurement samples.
- NumPy: classical diagonalization of the full and reduced Hamiltonians.

Methodology

1. Construct the Hamiltonian using Qiskit's `SparsePauliOp`.
2. Calculate the exact classical ground-state energy for reference.
3. Prepare and measure a quantum circuit using Qiskit's local `StatevectorSampler`.
4. Collect unique measured bitstrings to form the reduced computational-basis subspace.
5. Diagonalize the reduced Hamiltonian with NumPy.
6. Compare the sampled subspace with random subspaces of equal dimension.

Experiments and Interpretation

Experimental design

- Measurement shots: 16 to 4096.
- Repeat trials at each shot count.
- Record unique sampled states and estimated ground-state energies.
- Compare sampled and random subspaces against the exact reference.

What we investigate

- Does increasing the number of shots reveal more unique basis states?
- How does the reduced-space energy compare with the exact ground-state energy?
- How does the sampled subspace compare with a random subspace?

Expected interpretation: More shots can expand the sampled subspace, but finite-sample energy estimates need not improve monotonically.

Limitations and Conclusion

Important limitation

The circuit is initialized using the exact ground-state vector calculated classically. This is a controlled demonstration of sampling and subspace diagonalization, not a method for discovering an unknown ground state.

- The experiment uses a local statevector simulator, not quantum hardware.
- NumPy performs the reduced-subspace diagonalization.
- A random subspace is a simple baseline, not a comprehensive benchmark.
- No quantum advantage is claimed.

Conclusion: The notebook demonstrates how measurement outcomes can define a smaller subspace for classical ground-state energy estimation.

Declaration of Assistance and Resources

AI Tools: ChatGPT (OpenAI) assisted with project planning, explaining sample-based quantum diagonalization, organizing Qiskit code, and drafting the report and presentation.

Software and Computational Resources: Python and Qiskit were selected for Hamiltonian representation, quantum-circuit construction, and measurement sampling using a local simulator. NumPy was used for classical matrix operations and eigenvalue calculations. No quantum hardware execution or quantum advantage is claimed.

References and External Materials:
Qiskit documentation: `https://docs.quantum.ibm.com/`
Qiskit Fall Fest repository: `https://github.com/uiqc-club/qff26xUIQC`
These resources provided technical documentation and the context for the challenge.

Mentorship and Other Assistance: No external mentor or human technical collaborator is identified in this declaration. Any additional assistance received should be acknowledged before submission.

Limitations and Responsibility: The demonstration uses a ground-state vector calculated classically to initialize the circuit. It does not demonstrate independent quantum discovery of the ground state. The team is responsible for checking the code and reporting only results actually generated and verified.

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